Assessment Menu: Teacher Guide

The Circles Bistro · design notes, grading and answer key

Why students cannot just pick the easy ones

1. **Points are capped by category.** Students must complete exactly 4 / 4 / 2 dishes, so they cannot fill the total with Appetizers.
2. **Chef's Specials carry 40% of the grade** and Entrées another 40%. A student who only masters the basics tops out at 20 points.
3. **Skill coverage rule.** Their 10 picks must include at least one dish from each skill group below, so every strand of the unit is assessed.
4. **Harder dishes reward process.** Each has a rubric that gives credit for correct steps, so a partly correct attempt at a harder problem earns more than a perfect easy one in the same time.

Skill group	What it covers	Dishes
MEASURE	Circumference, area, and the equation of a circle	A1, A2, E1, S1
ARC	Arc length and sector area	A3, A4, E2, S2
ANGLE	Central and inscribed angles	A5, E3, E4, S3
LINES	Chords, tangents, and secants	A6, E5, E6, S4

If a student only does…	Maximum
Only Appetizers (4 × 5)	20
Appetizers + Entrées (4 × 5 + 4 × 10)	60
Full meal (adds 2 Chef's Specials)	100

Grading and options

- Grade only the marked dishes using the rubrics on the next pages (partial credit is built in). Total possible: **100**. If a student marks more than the required number, grade the first in each category.
- Check the skill rule on the order form before totaling. If a skill group is missing, apply your own policy (for example, the dish that broke the rule scores 0, or the student may swap one dish).
- Optional "second helping": allow one extra dish per category that replaces the student's lowest score in that category.
- Look at *which* dishes students chose. Students who select only the easiest dish in each skill group may need a conference.

Student	Appetizers (of 20)	Entrées (of 40)	Specials (of 40)	Skills OK?	Total

Answer Key: Appetizers (5 pts each)

Worked solutions and rubrics. Graphs: draw on the number line; endpoints as noted.

Dish	Problem	Solution	Rubric
A1 MEASURE	A circle has radius 7 cm. Find its circumference and its area. Leave answers in terms of π.	**Answer:** C = 14π cm; A = 49π cm² Circumference: C = 2πr = 2π(7) = 14π cm Area: A = πr² = π(7²) = 49π cm²	2 pts: Circumference 14π cm 3 pts: Area 49π cm² (square units)
A2 MEASURE	A circle has circumference 20π in. Find its radius and its area, in terms of π.	**Answer:** r = 10 in; A = 100π in² 2πr = 20π, so r = 10 in A = πr² = π(10²) = 100π in²	2 pts: Radius 10 in from 2πr = 20π 3 pts: Area 100π in²
A3 ARC	Find the length of a 90° arc of a circle with radius 12 m. Leave your answer in terms of π.	**Answer:** 6π m Arc length = $\frac{90}{360} \cdot 2\pi(12)$ = $\frac{1}{4} \cdot 24\pi = 6\pi$ m	2 pts: Uses the fraction 90 over 360 (or 1 over 4) of the circumference 3 pts: Correct arc length 6π m
A4 ARC	Find the area of a sector with central angle 60° in a circle of radius 6 cm, in terms of π.	**Answer:** 6π cm² Sector area = $\frac{60}{360} \cdot \pi(6^2)$ = $\frac{1}{6} \cdot 36\pi = 6\pi$ cm²	2 pts: Uses the fraction 60 over 360 of the circle area 3 pts: Correct area 6π cm²
A5 ANGLE	Central angle ∠AOB measures 84°, and inscribed angle ∠ACB intercepts the same arc AB. Find m∠ACB.	**Answer:** 42° A central angle equals its intercepted arc, so arc AB = 84°. An inscribed angle is **half** its intercepted arc: m∠ACB = 42°.	2 pts: Arc AB = 84° 3 pts: Inscribed angle is half: 42°
A6 LINES	Point P is outside a circle with center O. Segment PT is tangent to the circle at T, OT = 5 and PT = 12. Find OP.	**Answer:** OP = 13 A radius drawn to the point of tangency is perpendicular to the tangent, so △OTP has a right angle at T. $OP^2 = 5^2 + 12^2 = 169$ OP = 13	2 pts: Recognizes the right angle at T (radius ⊥ tangent) 3 pts: Pythagorean Theorem gives OP = 13

Answer Key: Entrées (10 pts each)

Worked solutions and rubrics. Graphs: draw on the number line; endpoints as noted.

Dish	Problem	Solution	Rubric
E1 MEASURE	A circle has center (−2, 3) and passes through the point (1, 7). Write the equation of the circle in standard form and find its area in terms of π.	**Answer:** $(x + 2)^2 + (y - 3)^2 = 25$; area = 25π Radius = distance from center to (1, 7): $\sqrt{((1 - (-2))^2 + (7 - 3)^2)} = \sqrt{(9 + 16)} = 5$ Standard form $(x - h)^2 + (y - k)^2 = r^2$: $(x + 2)^2 + (y - 3)^2 = 25$ Area = $\pi r^2 = 25\pi$	3 pts: Finds r = 5 with the distance formula 3 pts: Correct signs in (x + 2) and (y − 3) 2 pts: Right side is $r^2 = 25$ (not 5) 2 pts: Area 25π
E2 ARC	A sector of a circle has area 15π cm² and central angle 150°. Find the radius of the circle and the length of the sector's arc, in terms of π.	**Answer:** r = 6 cm; arc length = 5π cm $\frac{150}{360} = \frac{5}{12}$, so $\frac{5}{12} \cdot \pi r^2 = 15\pi$ $r^2 = 36$, so r = 6 cm Arc length = $\frac{5}{12} \cdot 2\pi(6) = 5\pi$ cm	3 pts: Sets up fraction times $\pi r^2 = 15\pi$ 3 pts: Solves r = 6 cm 2 pts: Arc length setup with the same fraction 2 pts: Arc length 5π cm
E3 ANGLE	Quadrilateral ABCD is inscribed in a circle. m∠A = (3x + 10)°, m∠C = (5x − 30)°, and m∠B = (4x)°. Find x, then find m∠A, m∠C and m∠D.	**Answer:** x = 25; ∠A = 85°, ∠C = 95°, ∠D = 80° Opposite angles of an inscribed quadrilateral are supplementary: (3x + 10) + (5x − 30) = 180 8x − 20 = 180, so x = 25 ∠A = 85°, ∠C = 95°, and ∠B = 4(25) = 100° ∠D = 180° − 100° = 80°	3 pts: States opposite angles are supplementary (sum 180°) 2 pts: Equation (3x + 10) + (5x − 30) = 180 2 pts: x = 25 1 pts: ∠A = 85° and ∠C = 95° 2 pts: ∠D = 80° using ∠B = 100°
E4 ANGLE	Mia says: "Inscribed angle ∠PQR intercepts arc PR, which measures 140°, so m∠PQR = 140°." Find her mistake, give the correct m∠PQR, and find the central angle ∠POR (O is the center).	**Answer:** An inscribed angle is half its intercepted arc: m∠PQR = 70°. The central angle m∠POR = 140°. Mia treated the inscribed angle as if it equaled the arc. That rule is for a **central** angle. Inscribed angle = half the arc: 140° ÷ 2 = 70°. Central angle ∠POR equals its arc: 140°.	4 pts: Identifies mistake: an inscribed angle is half its intercepted arc 3 pts: Correct m∠PQR = 70° 3 pts: Central angle equals its arc: 140°
E5 LINES	Chords AB and CD intersect at point P inside a circle. AP = 2 and PB = 16. If CP = x and PD = 2x, find x and the length of CD.	**Answer:** x = 4; CD = 12 Intersecting chords: AP · PB = CP · PD 2 · 16 = x · 2x, so $2x^2 = 32$ and $x^2 = 16$ x = 4 (a length cannot be −4) CD = x + 2x = 3x = 12	3 pts: Uses AP · PB = CP · PD 2 pts: Equation $2x^2 = 32$ 3 pts: x = 4 and rejects −4 2 pts: CD = 12
E6 LINES	From point P outside a circle, tangent segment PT = 6. A secant from P meets the circle at A (nearer) and B (farther), with PA = 4. Dev wrote: "$6^2 = 4 + PB$, so PB = 32." Find his mistake, then find the correct PB and AB.	**Answer:** He added instead of multiplying: $PT^2 = PA \cdot PB$. PB = 9 and AB = 5. Tangent-secant rule: (tangent)² = (whole secant) · (outside part), so $PT^2 = PA \cdot PB$. 36 = 4 · PB, so PB = 9. AB = PB − PA = 9 − 4 = 5.	3 pts: Identifies mistake: should multiply, $PT^2 = PA \cdot PB$ 3 pts: Equation 36 = 4 · PB, PB = 9 2 pts: AB = PB − PA 2 pts: AB = 5

Answer Key: Chef’s Specials (20 pts each)

Worked solutions and rubrics. Graphs: draw on the number line; endpoints as noted.

Dish	Problem	Solution	Rubric
S1 MEASURE	A circular garden has radius 10 m. A circular pond of radius 4 m sits at its center, and a gravel path 2 m wide surrounds the garden. **(a)** Find the planted area. **(b)** Find the path area. **(c)** Gravel costs \$3 per m²; find the cost to the nearest dollar. **(d)** Eli says the path area is $\pi(2)^2 = 4\pi$. Explain his error.	**Answer:** (a) 84π m² (b) 44π m² (c) about \$415 (d) The path is a ring, not a circle of radius 2. (a) Planted = garden − pond = $\pi(10^2) - \pi(4^2) = 100\pi - 16\pi = 84\pi$ m² (b) The path is a ring with outer radius 10 + 2 = 12: $\pi(12^2) - \pi(10^2) = 144\pi - 100\pi = 44\pi$ m² (c) $44\pi \times 3 = 132\pi \approx \$414.69$, so about \$415 (d) Eli used a circle of radius 2. The path is the area between the outer circle (radius 12) and the garden circle (radius 10), so you must subtract the inner circle’s area.	5 pts: (a) $100\pi - 16\pi = 84\pi$ m² 6 pts: (b) Outer radius 12 (3 pts); $144\pi - 100\pi = 44\pi$ m² (3 pts) 4 pts: (c) $132\pi \approx \$415$ 5 pts: (d) Explains the path is a ring and the inner circle must be subtracted
S2 ARC	A sector has central angle 72° and arc length 4π cm. **(a)** What fraction of the circle is the sector? **(b)** Find the radius. **(c)** Find the sector’s area. **(d)** Find the sector’s perimeter (arc plus two radii), in terms of π.	**Answer:** (a) $\frac{1}{5}$ (b) 10 cm (c) 20π cm² (d) $(4\pi + 20)$ cm (a) $\frac{72}{360} = \frac{1}{5}$ (b) $\frac{1}{5} \cdot 2\pi r = 4\pi$, so $2\pi r = 20\pi$ and $r = 10$ cm (c) Area $= \frac{1}{5} \cdot \pi(10^2) = 20\pi$ cm² (d) Perimeter = arc + 2r = $4\pi + 20$ cm	3 pts: (a) Fraction 1 over 5 6 pts: (b) Sets up fraction times circumference = 4π (3 pts); r = 10 (3 pts) 5 pts: (c) Area 20π cm² 6 pts: (d) Adds two radii (3 pts); perimeter $4\pi + 20$ (3 pts)
S3 ANGLE	Triangle ABC is inscribed in a circle and AB is a diameter. $m\angle A = (2x + 4)°$ and $m\angle B = (3x - 9)°$. **(a)** Explain why $m\angle C = 90°$. **(b)** Find x and both angles. **(c)** Find the measures of arc BC (not containing A) and arc AC (not containing B).	**Answer:** (a) ∠C intercepts a semicircle (180°), so it is 90°. (b) x = 19; ∠A = 42°, ∠B = 48° (c) arc BC = 84°, arc AC = 96° (a) AB is a diameter, so ∠C intercepts a 180° arc. An inscribed angle is half its arc: 90°. (b) Angles of a triangle: $(2x + 4) + (3x - 9) + 90 = 180$, so $5x = 95$ and $x = 19$. ∠A = 42°, ∠B = 48°. (c) Each arc is twice the inscribed angle that intercepts it: arc BC = 2(42°) = 84°, arc AC = 2(48°) = 96°. Check: 84° + 96° = 180°, a semicircle.	4 pts: (a) Diameter gives a 180° arc and an inscribed angle is half of it 4 pts: (b) Equation $(2x + 4) + (3x - 9) = 90$ 2 pts: (b) x = 19 2 pts: (b) ∠A = 42° and ∠B = 48° 4 pts: (c) Arc BC = 84° 4 pts: (c) Arc AC = 96°
S4 LINES	From point P outside a circle with center O and radius 12, tangent segments PA and PB touch the circle at A and B. PA = $3x - 2$ and PB = $x + 10$. **(a)** Find x and PA. **(b)** Find OP. **(c)** Find the area of quadrilateral OAPB. **(d)**	**Answer:** (a) x = 6, PA = 16 (b) OP = 20 (c) 192 square units (d) A radius meets a tangent at 90°. (a) Tangent segments from one outside point are equal: $3x - 2 = x + 10$, so x = 6 and PA = 16. (b) △OAP is right at A: $OP^2 = 12^2 + 16^2 = 400$, so OP = 20. (c) OAPB is two congruent right	3 pts: (a) Tangent segments equal: $3x - 2 = x + 10$ 3 pts: (a) x = 6 and PA = 16 4 pts: (b) Right triangle, $OP^2 = 12^2 + 16^2$, OP = 20 5 pts: (c) Two congruent right triangles; area 192

	Explain why ∠OAP is a right angle.	triangles: $2 \cdot \frac{1}{2} \cdot 12 \cdot 16 = 192$ (d) A radius drawn to the point of tangency is perpendicular to the tangent line.	5 pts: (d) Radius to the point of tangency is perpendicular to the tangent